

JACOB PRINZ

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EDUCATION

University of Maryland, College Park

August 2020 - (planned) 2026

PhD in Computer Science

Title: Typed Embeddings for Metaprogramming

Advisor: Leonidas Lampropoulos

University of Maryland, College Park

August 2016 - May 2020

B.S. in Mathematics

WORK EXPERIENCE

Programming Language Lab - University of Maryland

Aug 2020 - Present

- Graduate Research Assistant. Advisor: Leonidas Lampropoulos

Developing proof assistants and proof automation in Lean, Rocq, and Agda. Developed formalization techniques in dependent type theory. Developed property based testing technique and incorporated into widely used Rocq library called QuickChick.

REU - Indiana University

Jun 2018 - Aug 2018

- Research Intern. Advisor: Noah Snyder

Studied Homotopy Type Theory with Agda proof assistant. Calculated homotopy groups of spheres in Homotopy Type Theory.

Independent Study - University of Maryland

2016 - 2020

- Research Intern. Advisor: William Gasarch

Coauthored a book on a combinatorics problem, contributing two theorems and an integer programming algorithm.

Losert Lab, University of Maryland

Sep 2018 - Dec 2018

- Research Intern. Advisor: Anton Peshkov

Analyzed large dataset with C++ library for a particle physics experiment. Discovered a bug in a particle physics library.

High Energy Physics Group - Fermilab/University of Maryland

Jun 2017 - Aug 2017

- Research Intern. Advisor: Sarah Eno

Wrote data analysis code for a particle physics experiment. Discovered a bug in a particle physics library.

Catt Labs

Jun 2014 - Aug 2015

- Software Developer, Intern

Developed a webserver and web applications to disseminate traffic data in Java, JavaScript, and SQL.

PUBLICATIONS

- A Monad for Nonconstructivity** Preprint
Jacob Prinz and Leonidas Lampropoulos
- Pantograph: A Fluid and Typed Structure Editor** POPL 2025
Jacob Prinz, Henry Blanchette, and Leonidas Lampropoulos
- Merging Inductive Relations** PLDI 2023
Jacob Prinz and Leonidas Lampropoulos
- Deeper Shallow Embeddings** ITP 2022
Jacob Prinz and Leonidas Lampropoulos

BOOKS

- Mathematical Muffin Morsels: Nobody Wants a Small Piece** May 2020
William Gasarch, Erik Metz, **Jacob Prinz**, and Daniel Smolyak

TALKS

- Merging Inductive Relations NJPLS May 2023
Jacob Prinz
- Invited Talk on Deeper Shallow Embeddings Portland PLV Seminar Apr 2023
Jacob Prinz
- Zypr: A New Approach to Structural Editing NJPLS Oct 2022
Henry Blanchette, Jacob Prinz
- Deeper Shallow Embeddings NJPLS May 2022
Jacob Prinz

TEACHING

- UMD CMSC631 **Program Analysis and Understanding** Spring 2025
Teaching Assistant
Professor: Leonidas Lampropoulos
- UMD CMSC433 **Programming Language Technologies and Paradigms** Fall 2021
Teaching Assistant
Professor: Anwar Mamat
- UMD CMSC456 **Introduction to Cryptography** Fall 2018
Teaching Assistant
Professor: William Gasarch
- UMD MATH299T **Homotopy Type Theory** Spring 2019
Student Initiated Course Lecturer
Student Initiated Course Supervisor: Sarah Yeakel